

ASSIGNMENT-VII

1. Write a program to perform **linear search** on an array.

Sample Input:

```
Array: 10 20 30 40 50  
Key: 30
```

Sample Output:

```
Element found at position 3
```

2. Write a program to sort elements using **bubble sort**.

Sample Input:

```
Array: 5 2 9 1 6
```

Sample Output:

```
Sorted Array: 1 2 5 6 9
```

3. Write a program to:
 - Sort an array using insertion sort technique
 - Then search an element using binary search

Sample Input:

```
Array: 12 4 7 1 9  
Key: 7
```

Sample Output:

```
Sorted Array: 1 4 7 9 12  
Element found at position 3
```

4. Write a program to sort an array using **Quick Sort**.

Sample Input:

```
Array: 10 7 8 9 1 5
```

Sample Output:

```
Sorted Array: 1 5 7 8 9 10
```

5. Write a program to implement **Merge Sort**

Sample Input:

Array: 38 27 43 3 9 82 10

Sample Output:

Sorted Array: 3 9 10 27 38 43 82

6. Write a program to find the **kth smallest element** in an array.

Sample Input:

Array: 7 10 4 3 20 15

k = 3

Sample Output:

3rd smallest element: 7